
Seasonal odds for water-energy planning: AI-ready teleconnection data and a logistic-referenced residual Transformer

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Abstract

Weeks-to-years prediction of water for energy sits between weather forecasts and long-range projections: long enough that reservoir and generation decisions matter, short enough that year-to-year variability still dominates. In the Tennessee Valley Authority (TVA) region, winter wet/dry and cool/hot regimes shift hydropower inflows and thermoelectric cooling demand, so even categorical seasonal *odds* would be a useful planning product for reservoir and generation operations. The problem is hard. Labelled winters are few; month-by-month anomaly regression on teleconnections collapses toward climatology; and operational seasonal systems such as CFSv2 are only a weak *local* guide for categorical wet/dry outcomes. We therefore isolate a first experiment whose purpose is to test whether a documented large-scale ocean–atmosphere state can shift TVA winter odds at all, before any cascade to streamflow or energy models. All learners share one AI-ready pack of teleconnection indices and sea-surface temperature (SST) boxes. A multinomial logistic model is the skill reference. On a blocked test of 15 winters, logistic hit rate and Heidke skill for wet/dry are 0.67 and 0.43 (10/15 winters), versus 0.47 and 0.11 (7/15) for October-initialized CFSv2 on the same reanalysis labels. Fully connected, LSTM, and CNN models do not match that reference; a residual temporal Transformer (logistic path plus a constrained attention residual) matches it on both tasks, with a Brier-skill gain on wet/dry rather than a higher hit rate. Skill is mainly a winter signal and fades beyond about two seasons. The result is evidence that the seasonal-odds question is tractable—not a weeks-to-years hydrology system and not an operational TVA product.

1 Introduction

A challenging but meaningful problem. Water for energy is decided on a weeks-to-years horizon that neither tomorrow’s weather model nor a century-scale projection is built to serve. Hydropower and thermoelectric plants in the TVA region depend on water availability, streamflow, and river temperature. A wet versus dry winter changes reservoir inflows; a cool versus hot winter changes heating and cooling load. Those seasonal regimes are tied to large-scale ocean–atmosphere modes: ENSO for North American winter precipitation and temperature [5, 6], and the North Atlantic

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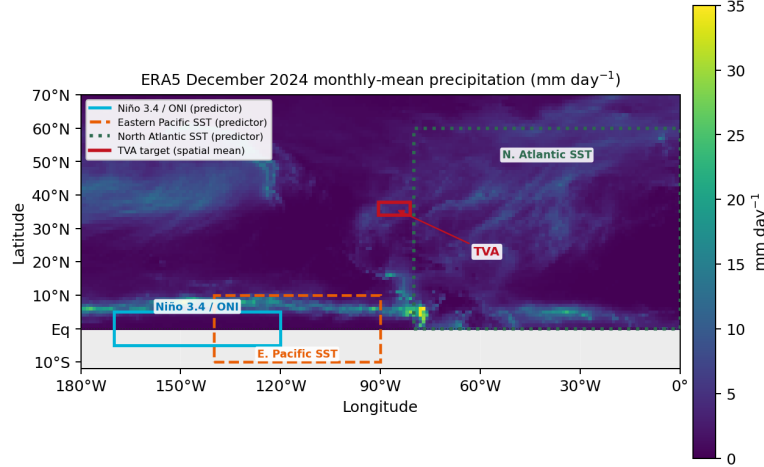


Figure 1: ERA5 December 2024 monthly-mean precipitation (mm day^{-1}) over the daily 1° analysis window, with the index and SST boxes used in this paper. Cyan: Niño 3.4 / ONI; dashed orange: eastern Pacific SST; dotted green: North Atlantic SST; red: TVA spatial-mean target. NAO and AO are atmospheric indices, not additional SST boxes. The southern half of Niño 3.4 lies south of the ERA5 cutoff at the equator (grey); index and SST-box time series still use the full conventional box.

Oscillation (NAO) and Arctic Oscillation (AO) for much of the eastern U.S. winter temperature variance [7, 8]. If public observations and machine learning can shift the *odds* of those regimes, a seasonal wet/dry outlook becomes a planning input for existing TVA reservoir and hydropower operations under year-to-year variability—valuable even though it is not yet a deployed forecast.

The same problem is scientifically hard. The usable sample is small once winters are blocked for honest evaluation. Pooling every calendar month as a continuous anomaly dilutes the seasonal signal, so mean-squared-error models look near climatology. Operational seasonal products such as CFSv2 [9], a member of the North American Multi-Model Ensemble (NMME) [10], need not be a strong *local* categorical baseline for TVA wet/dry verification. A full cascade from large-scale predictors to streamflow and energy-economy indicators cannot create skill that is absent in the index-to-seasonal-target step.

Purpose of this experiment. We therefore design a constrained first test: can autumn teleconnection indices and tropical Pacific SST, drawn from one curated pack, shift December–February (DJF) TVA wet/dry and cool/hot tercile odds? If that signal cannot be recovered with a transparent logistic model, later spatial, land-runoff, or energy models are unlikely to invent it. The experiment is an early step in a U.S. Department of Energy Genesis Mission project on water for energy; it is evidence-oriented on public data, not an operational TVA system.

How, and what we find. Figure 1 shows the geography: teleconnection indices and three OISST boxes as predictors; a TVA ERA5 spatial mean as the target, not map pixels [1–4]. All models see the same inputs. Logistic regression on September–November (SON) means is the skill reference; CFSv2 is operational context on the same ERA5 labels, not the bar a learning model must beat; a residual Transformer may only refine logistic. On 15 blocked test winters, logistic already exceeds climatology and this local CFSv2 comparison for wet/dry (10/15 versus 7/15 hits).

2 Related work

Teleconnection diagnostics have long linked tropical Pacific SST and North Atlantic pressure patterns to U.S. winter precipitation and temperature [5–8]. Operational dynamical seasonal prediction—CFSv2 and NMME—provides a public benchmark [9–11], but global ENSO skill does not automatically transfer to categorical wet/dry verification on a regional energy box. Machine learning has been applied to ENSO prediction, including deep networks for multi-year ENSO [12]. We instead score

TVA winter *odds* with a logistic reference so a Transformer is accepted only if it matches or improves that reference on the same curated inputs [13–15]. We train and score on ERA5 TVA means [1] and treat Daymet [17] as a later dual-truth path, not this paper’s target.

3 Data

Every model in this paper—logistic, fully connected (FC) network, LSTM, CNN, residual Transformer—and the CFSv2 categorical comparison use the same public streams: monthly teleconnection indices, including the Oceanic Niño Index (ONI) and Niño 3.4 [3, 4]; SST box anomalies from NOAA OISST [2]; and TVA-mean ERA5 anomalies of near-surface air temperature, precipitation, and mean sea-level pressure [1]. These streams are issued as a monthly labelled pack: one sample per forecast issue month ($N=486$ months, August 1982–January 2023) with a 12-month lookback. The experiments below do not train a second dataset; they aggregate the same series into seasonal means or terciles.

The predictand is ERA5-based, not Daymet. Daily 1° fields over the precipitation window of Fig. 1 are averaged over the TVA box, converted to monthly anomalies, then to DJF terciles (below-normal, near-normal, above-normal) with edges estimated on training winters only. Logistic regression uses SON *means* of the index/SST series; sequence models use the last $L=12$ months of the same series. Reserved dataset DOIs and code URLs appear in the camera-ready data statement.

4 Methods

Tasks. *Winter wet/dry* (primary): SON ONI and Niño 3.4 box anomaly predict the DJF TVA precipitation tercile (dry / near / wet). *Winter cool/hot*: SON NAO and AO predict the DJF near-surface temperature tercile (cool / near / hot).

Split. Train on DJF years ≤ 2007 ($n=25$); test on ≥ 2009 ($n=15$); hold out 2008. Neural models use a five-seed soft-vote ensemble on test.

Scores. Hit rate is the fraction of test winters assigned the correct tercile (chance $\approx 1/3$). Heidke skill measures categorical skill beyond chance ($0 \approx$ chance) [13, 15]. Brier skill score (BSS) is relative to climatological class frequencies [14, 15]; $BSS > 0$ improves on climatology as a probability forecast. These scores match a planner-facing *odds* product rather than pixel RMSE. We say a model *matches the logistic reference* when hit, Heidke, and BSS are all at least as large as logistic on that task.

Models. Multinomial logistic regression on SON means; an FC network on SON means; LSTM and CNN on $L=12$ monthly sequences; and a residual temporal Transformer [16]

$$\text{logits} = \text{logistic}(\text{SON mean}) + \text{softplus}(\alpha) \text{Transformer}(L=12). \quad (1)$$

With ~ 25 training winters, a pure sequence model must rediscover the SON-mean teleconnection from scratch and, in our trials, failed the wet/dry task. The residual form keeps the logistic path (warm-started on SON means) and lets attention learn only a constrained correction.

5 Results

Figure 2 reports the blocked test (exact numbers in Appendix Table 3). Panel (a) is *local operational context* on wet/dry, not a claim that a deep network outplays CFSv2. October-initialized CFSv2 ensemble-mean precipitation is mapped to the TVA box and scored as DJF terciles against the **same ERA5 wet/dry labels**. It exceeds the majority-class null (hit 0.47 versus 0.33) but remains weakly skillful categorically. Logistic regression on two autumn means is stronger (hit 0.67, Heidke 0.43): a 10/15 versus 7/15 difference on this pilot test. CFSv2 is therefore context, not the reference a learning model must beat.

Panel (b) compares architectures on those same predictors. FC clones logistic wet/dry hit and Heidke but not BSS. LSTM and CNN fail the wet/dry task at this sample size. Only the residual Transformer matches logistic on all three metrics for both tasks; on wet/dry it is tied on hit and Heidke

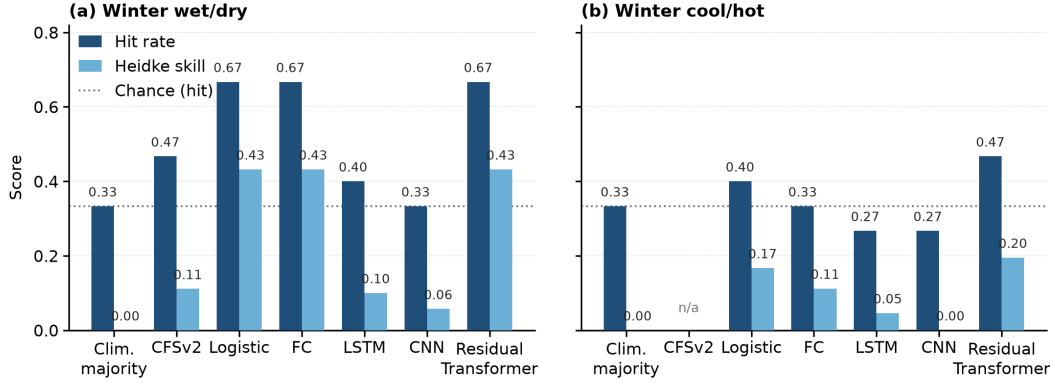


Figure 2: Blocked test ($n=15$) on identical AI-ready predictors; both panels use the same axes, bar width, and model order. (a) Winter wet/dry, including October-initialized CFSv2 scored on the same ERA5 labels (local context, not a full CFSv2 verification). (b) Winter cool/hot; CFSv2 was not scored on this task. Dotted line: chance hit (1/3). Logistic is the skill reference; the residual Transformer matches it on hit and Heidke for wet/dry.

and improves BSS (0.091 to 0.140). “Strongest” means strongest among this closed comparison, and the winning model still contains a logistic branch. Cool/hot remains modest (logistic hit 0.40; Transformer 0.47, one extra winter).

6 Discussion and limitations

Two extra checks bound the claim. Using the same winter-style indices on other meteorological seasons, spring and summer do not pass; autumn wet/dry is only a weak exploratory check (hit 0.40, negative Heidke; Appendix A). Lengthening lead while keeping winter wet/dry as the target, a one-season issue (SON→DJF) reproduces the main result; two seasons (JJA→DJF) is a weak check (hit 0.40, BSS +0.024); three seasons (MAM→DJF) fails. A residual Transformer does not match logistic at the two-season lead. Useful index horizon is therefore about one to two seasons for winter wet/dry, not three seasons and not all calendars.

The test set contains only 15 winters, so scores are a pilot demonstration rather than an operational skill claim; we do not report interval estimates. Cool/hot skill is modest; the water-energy narrative is primarily wet/dry. The residual Transformer is not trained from a blank slate. CFSv2 is evaluated here for TVA-box categorical wet/dry odds, not as a global seasonal verification package [11]. This work does not yet forecast streamflow, river temperature, or energy-economy outcomes; those later steps should reuse the same odds language once daily or land targets exist.

7 Conclusion

Weeks-to-years water-for-energy planning needs seasonal TVA information that is difficult but, on AI-ready teleconnection data, tractable as winter odds. Logistic regression already exceeds climatology and this local CFSv2 categorical comparison for wet/dry; a residual Transformer is the strongest model we compared only in the sense that it matches that logistic reference, with logistic retained as the skill floor. That is evidence for a ground-up water-energy planning step—not a weeks-to-years hydrology system, not an image downscaler, and not a deployed TVA product. Next data work is a daily or weekly pack, not further architecture search on the monthly index tensor.

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Data and code. Observation pack (reserved DOI <https://doi.org/10.13139/ORNLNCCS/3398576>) and monthly labelled training pack (reserved DOI <https://doi.org/10.13139/ORNLNCCS/3409936>). Staging: https://github.com/daliwang/water4energy_data, https://github.com/daliwang/water4energy_trainingData. Models: https://github.com/daliwang/water4energy_model.

Competing interests. The authors declare no competing interests.

A Calendar transfer, longer leads, and numeric scoreboard

Table 1 uses the same logistic recipe with a strict prior-season lag (train season-years ≤ 2007 , test ≥ 2009). Table 2 keeps winter wet/dry and issues the forecast one, two, or three seasons ahead. Table 3 lists the blocked-test numbers plotted in Fig. 2.

Table 1: Logistic skill by meteorological season (strict prior-season predictors). A pass requires $BSS > 0$ and ($Heidke > 0$ or hit above the majority class). Autumn wet/dry meets this rule only weakly (negative Heidke) and is treated as exploratory in the main text.

Season	Task	Hit	Heidke	BSS	Pass
DJF (winter)	wet/dry	0.667	0.432	0.091	yes
DJF (winter)	cool/hot	0.400	0.167	0.009	yes
MAM (spring)	wet/dry	0.333	-0.304	-0.207	no
MAM (spring)	cool/hot	0.133	-0.016	-0.122	no
JJA (summer)	wet/dry	0.400	0.075	-0.011	no
JJA (summer)	cool/hot	0.333	0.063	-0.091	no
SON (autumn)	wet/dry	0.400	-0.125	0.016	yes*
SON (autumn)	cool/hot	0.400	-0.063	-0.295	no

*Exploratory: $BSS > 0$ but $Heidke < 0$.

Table 2: Logistic winter wet/dry skill versus issue lead (cool/hot at two seasons shown for contrast).

Issue $\rightarrow$ target	Hit	Heidke	BSS	Pass
SON $\rightarrow$ DJF (one season)	0.667	0.432	0.091	yes
JJA $\rightarrow$ DJF (two seasons)	0.400	0.135	0.024	yes
MAM $\rightarrow$ DJF (three seasons)	0.133	-0.121	-0.230	no
JJA $\rightarrow$ DJF cool/hot	0.267	-0.071	-0.084	no

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Table 3: Blocked test scores on identical AI-ready predictors. CFSv2 is shown for wet/dry as operational context only.

Task	Model	Hit	Heidke	BSS	vs logistic
Wet/dry	CFSv2 Oct→DJF	0.467	0.111	—	below
Wet/dry	Logistic	0.667	0.432	0.091	reference
Wet/dry	FC	0.667	0.432	0.051	below (BSS)
Wet/dry	LSTM	0.400	0.100	−0.016	fail
Wet/dry	CNN	0.333	0.057	−0.038	fail
Wet/dry	Residual Transformer	0.667	0.432	0.140	matches
Cool/hot	Logistic	0.400	0.167	0.009	reference
Cool/hot	FC	0.333	0.112	0.043	partial
Cool/hot	LSTM	0.267	0.046	0.037	partial
Cool/hot	CNN	0.267	0.000	0.022	partial
Cool/hot	Residual Transformer	0.467	0.195	0.050	matches

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